ARISH SINGH

AI / ML Engineer — LLM & RAG Systems (Student)

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LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate specialising in Artificial Intelligence, with hands-on experience in machine learning, NLP, LLM integration, and full-stack development. Demonstrated ability to build end-to-end AI systems including RAG pipelines, vector database setups, and web applications. Eager to contribute technical skills and a problem-solving mindset in an entry-level role while continuing to grow professionally.

TECHNICAL SKILLS

Programming Languages:	Python (Strong), Java, SQL
AI / ML:	Machine Learning, Deep Learning, NLP, RAG, Embedding Generation, Text Chunking
LLM Frameworks:	Hugging Face Transformers, LangChain, Ollama, llama.cpp (Learning)
Vector Databases:	Qdrant, ChromaDB, FAISS, Weaviate (Learning)
Libraries:	Pandas, NumPy, Matplotlib, Scikit-learn
Web Dev:	HTML, CSS, JavaScript, Flask, React.js, FastAPI
Databases:	SQL, MongoDB
Tools & Infra:	Git, GitHub, Docker (Basic), Jupyter Notebook, VS Code
Soft Skills:	Teamwork, Problem-Solving, Time Management, Adaptability

EDUCATION

B.Tech in Computer Science – AI Specialisation 2023 – 2027

Parul Institute of Engineering & Technology, Vadodara

Higher Secondary (Class XII) – GHSEB 2022 – 2023

Shree Shreyas Vidyalaya | Mathematics, Physics, Chemistry & Computer Science

ACADEMIC PROJECTS

AI-Based Voice Interview System

- Built a real-time AI-powered interview system using MERN stack with resume parsing and dynamic question generation.
- Implemented NLP-based resume analysis to extract skills and tailor interview questions based on candidate profile.
- Integrated speech processing (STT/TTS) for seamless voice interaction and automated NLP-based response evaluation with real-time feedback.

InferIQ – AI-Powered Data Cleaning & Model Recommendation System

- Built an end-to-end AI tool automating ML data cleaning and model selection using Ollama LLM, Python FastAPI, and React — reducing model selection time by 3× and improving accuracy by up to 38%.
- Developed a React.js dashboard with Recharts visualizations for before/after data cleaning comparison and multi-model performance metrics (Accuracy, F1, RMSE).

RAG Pipeline – Document Q&A (Learning Project)

- Explored document ingestion pipeline to chunk and embed text (PDFs, DOCX) into a local vector store using open-source embedding models.
- Experimented with querying embedded data via local LLMs (Ollama/LLaMA/Mistral) to simulate Retrieval Augmented Generation.
- Practiced setting up and configuring vector DBs (ChromaDB, Qdrant) and designing simple API interfaces for querying via LLMs.
- Documented architecture flow, data pipeline, and configuration details for the project.

CERTIFICATIONS & ACHIEVEMENTS

- Participated in Guidewire DEVTrails University Hackathon 2026.

- Web Development – Hands-on training in front-end (HTML, CSS, JavaScript) and back-end development with SQL and MongoDB.
- CODE-X Participant – Gujarat's largest technical festival (Projections).

EXTRACURRICULAR ACTIVITIES

- Attended hands-on workshop on Generative AI organised by the CSE – AI & ML Department.
- Actively self-studying LLM frameworks (LangChain, Hugging Face, Ollama) and vector DB integration for RAG systems.
- Exploring embedding generation, cosine similarity, and text chunking strategies through self-practice and open-source projects.